MATHEMATICS

Applications and interpretation

Higher Level | Set A | Paper 3

Challenge mock examination

Time allowed: 1 hour 15 minutes Maximum mark: 55

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

• A graphic display calculator and a clean Mathematics: applications and interpretation formula booklet are required.

• Answer all questions. Show working and explain mathematical reasoning in context.

• This paper contains two compulsory extended problem-solving investigations.

• Give exact answers where appropriate. Otherwise give numerical answers to three significant figures, unless a question specifies a different accuracy.

• Keep unrounded values in subsequent calculations. State units where relevant.

• Diagrams are not necessarily drawn to scale. Use the stated data and conditions.

• The number in square brackets gives the maximum mark for a part.

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Question 1 | Question paper

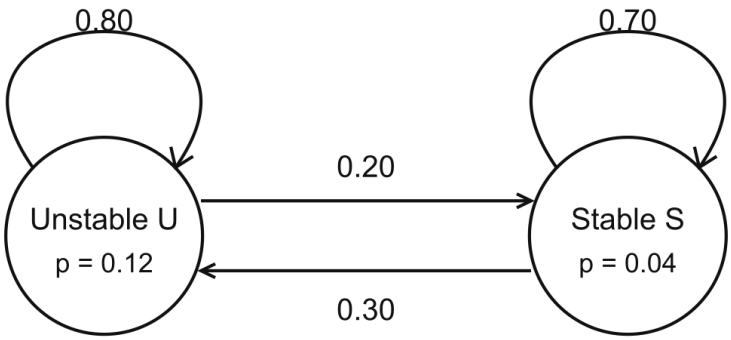
Question 1 [Maximum mark: 27]

A factory models each day's production as being in one of two states: unstable U or stable S. The state changes once per day according to the transition diagram. Initially, on day 0, the factory is known to be stable.

On each day an inspector takes 40 seals. Conditional on that day's state, the seals are independent and each seal is defective with probability 0.12 in state U, or 0.04 in state S. Let X be the number of defective seals in the sample. The sample is too small to change the state-transition probabilities.

The inspector tests $H_0 : p = 0.04$ against $H_1 : p > 0.04$ at the 5% significance level, raising an alert when $X \geq c$. Choose the smallest integer c for which the Type I error probability is less than 0.05.

Let q_n be the probability that the factory is unstable on day n, and let a_n be the probability of an alert on day n. Use unrounded probabilities in subsequent calculations.



State changes once per day; seal probabilities are conditional on that day's state.

(a) Find c and the actual Type I error probability. Show why the next smaller value of c is unsuitable. [4]

(b) Find the Type II error probability when $p = 0.12$. Interpret this error in the factory context. [3]

(c) Using column states ordered (U, S), write down the transition matrix T. Find q_2. [3]

Question 1 | Question paper

(d) By considering $q_{n+1} - 0.6$, obtain an explicit formula for q_n valid for every integer $n \geq 0$. Show the reasoning leading to your formula. [5]

(e) In the long run, an alert is observed. Find the conditional probability that the factory is unstable on that day. [4]

(f) Find the smallest day number n for which $|a_n - \lim_{k\to\infty} a_k| < 0.001$. [3]

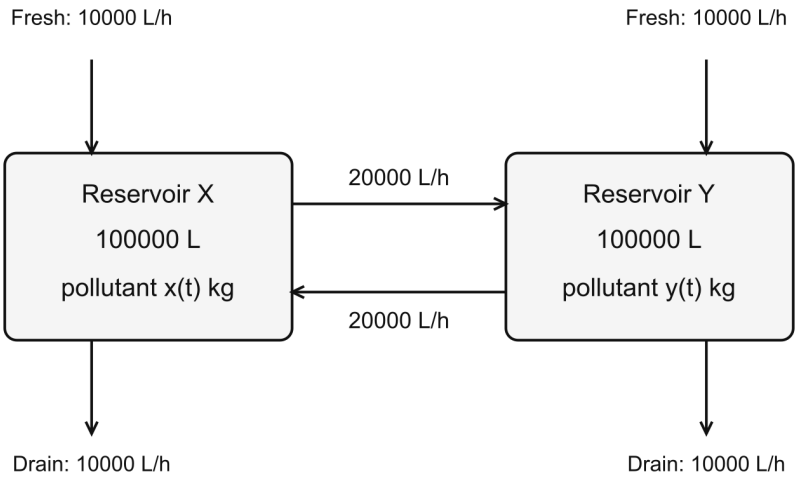
(g) For a more general factory, suppose $\mathrm{P}(U \to S) = \kappa u$ and $\mathrm{P}(S \to U) = \kappa v$, where $u, v > 0$ and $0 < \kappa(u + v) < 1$. The two conditional seal distributions and alert rule are unchanged. Show that the limiting alert probability is independent of κ. Explain how increasing κ affects the approach to this limit from a fixed initial state distribution. [5]

Question 2 [Maximum mark: 28]

Two reservoirs X and Y each contain $100\,000$ litres of water. In each direction, water is pumped between the reservoirs at $20\,000$ litres per hour. Each reservoir also receives pollutant-free water at $10\,000$ litres per hour and discharges water to a drain at the same rate. Each reservoir is perfectly mixed, and its volume stays constant.

At time $t = 0$, reservoir X contains 30 kg of pollutant and reservoir Y contains none. No further pollutant enters the system. Let $x(t)$ and $y(t)$ be the pollutant masses, in kg, at time t hours. Pollutant leaves each reservoir at its current concentration.

The diagram shows the flow rates. Take 1 kg to be 10^6 mg.



Each reservoir is kept at a constant volume and is perfectly mixed.

(a) Obtain the coupled differential equations for x and y, and express them as a matrix equation. [4]

(b) Use the eigenvalues and eigenvectors of your matrix to find exact expressions for $x(t)$ and $y(t)$. [6]

(c) Find the time at which reservoir Y contains the most pollutant, and find this maximum mass. Justify that it is a global maximum for $t \geq 0$. [4]

Question 2 | Question paper

(d) Both reservoirs are classified as safe when each concentration is at most 10 mg per litre. Find the earliest time at which both are safe, explaining which reservoir determines this time. [3]

(e) Apply Euler's method with step length $h = 1$ hour for two steps. Compare the resulting $y(2)$ with the exact value. A technician claims that an Euler model is accurate whenever it keeps both masses nonnegative. Evaluate this claim using your results. [3]

(f) Now replace each exchange flow of $20\,000$ litres per hour by an arbitrary rate $Q > 0$, leaving both fresh-water and drain rates unchanged. Find the half-life of the total pollutant mass. Does increasing Q make the total pollutant disappear faster? Justify your answer. [3]

(g) Return to the original exchange flow. Set $S_n = x_n + y_n$ and $D_n = x_n - y_n$ for an Euler approximation with step length $h > 0$. Derive recurrences for S_n and D_n. Determine the largest h that guarantees, for every initial state $x_0 \geq y_0 \geq 0$, that both masses stay nonnegative and the ordering $x_n \geq y_n$ is preserved at every step. Explain why this guarantee still does not establish accuracy. [5]

End of question paper

MATHEMATICS

Applications and interpretation

Higher Level | Set A | Paper 3

Worked solution book

Complete this paper before consulting the worked solutions. Each part has method (M), accuracy (A) and reasoning (R) points. The number following a code is its mark value. Award only the points supported by the candidate's retained working.

Accept mathematically equivalent valid methods. Numerical answers may use technology, but should be supported by equations, test definitions, bounds, labelled graphs or other appropriate mathematical working. Retain unrounded intermediate values.

Use the portal to enter a whole-number self-assessed mark for each part. These are independent practice papers; the self-assessed report does not calculate an official IB grade.

Question 1 | Worked solutions

Question 1 [Maximum mark: 27]

(a) [4 marks]

M1 Under H_0, use $X \sim \text{Bin}(40, 0.04)$ and upper-tail probabilities.

A1 $\text{P}(X \geq 4) = 0.0748372580 > 0.05$.

A1 $\text{P}(X \geq 5) = 0.0210222967 < 0.05$.

A1 Thus $c = 5$, with actual Type I probability $\alpha = 0.0210$ (three significant figures).

(b) [3 marks]

M1 Use $X \sim \text{Bin}(40, 0.12)$ and failure to reject, $X \leq 4$.

A1 $\beta = \text{P}(X \leq 4) = 0.4669109648 \approx 0.467$.

R1 The inspector fails to raise an alert although the factory is in its unstable, higher-defect state.

(c) [3 marks]

A1 $T = \begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix}$, with column entries representing transitions from that column state.

M1 Apply T^2 to $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$, or use $q_1 = 0.3$ and $q_2 = 0.8q_1 + 0.3(1 - q_1)$.

A1 $q_2 = 0.45$.

(d) [5 marks]

M1 $q_{n+1} = 0.8q_n + 0.3(1 - q_n) = 0.3 + 0.5q_n$.

A1 The stationary value solves $q = 0.3 + 0.5q$, so $q = 0.6$.

M1 $q_{n+1} - 0.6 = 0.5(q_n - 0.6)$, a geometric sequence of differences.

A1 $q_n - 0.6 = (q_0 - 0.6)(0.5)^n$.

A1 Since $q_0 = 0$, $q_n = 0.6(1 - 2^{-n})$, including $n = 0$.

(e) [4 marks]

M1 The limiting alert probability is $a_\infty = 0.6(1-\beta) + 0.4\alpha$.

A1 $a_\infty = 0.3282623398$.

M1 $\mathrm{P}(U \mid \text{alert}) = 0.6(1-\beta)/a_\infty$.

A1 $0.9743835412 \approx 0.974$.

(f) [3 marks]

M1 $a_n = \alpha + (1-\beta-\alpha)q_n$, so $|a_n - a_\infty| = 0.6(1-\beta-\alpha)2^{-n}$.

A1 Solving the inequality gives $n > 8.2632\ldots$, hence $n = 9$.

R1 At $n = 8$ the difference is $0.00120016\ldots$; at $n = 9$ it is $0.000600078\ldots$, verifying the first integer day.

(g) [5 marks]

M1 The general recurrence is $q_{n+1} = \kappa v + [1 - \kappa(u+v)]q_n$.

A1 The limiting unstable probability is $\pi = v/(u+v)$.

M1 The limiting alert probability is $\alpha + (1-\beta-\alpha)v/(u+v)$.

A1 This expression contains no κ, so the limit is unchanged.

R1 $q_n - \pi = [1-\kappa(u+v)]^n(q_0 - \pi)$. The factor lies in $(0,1)$ and decreases as κ increases, so convergence is faster (or the process is already stationary).

Question 2 | Worked solutions

Question 2 [Maximum mark: 28]

(a) [4 marks]

M1 Concentration is mass divided by 100000 L; total outflow from each reservoir is 30000 L/h.

M1 Pollutant inflow from the other reservoir is 0.2 times its pollutant mass per hour.

A1 $x\prime = -0.3x + 0.2y$, $y\prime = 0.2x - 0.3y$.

A1 $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} -0.3 & 0.2 \\ 0.2 & -0.3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

(b) [6 marks]

M1 Use the characteristic equation $(-0.3 - \lambda)^2 - 0.04 = 0$.

A2 Eigenvalues are -0.1 and -0.5.

A1 Corresponding eigenvectors may be taken as

$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

and

$\begin{pmatrix} 1 \\ -1 \end{pmatrix}$

M1 $\begin{pmatrix} x \\ y \end{pmatrix} = C_1 e^{-0.1t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + C_2 e^{-0.5t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

; apply $x(0) = 30$, $y(0) = 0$ to get $C_1 = C_2 = 15$.

A1 $x(t) = 15(e^{-t/10} + e^{-t/2})$, $y(t) = 15(e^{-t/10} - e^{-t/2})$.

(c) [4 marks]

M1 $y\prime = -1.5e^{-t/10} + 7.5e^{-t/2} = 0$.

A1 $e^{0.4t} = 5$, hence $t = (5/2)\ln 5 = 4.023594781\ldots$ h.

R1 $y\prime$ changes from positive to negative at this sole stationary point; $y(0) = 0$ and $y(t) \to 0$, so it is the global maximum.

A1 $y_{\max} = 8.024883660\ldots \approx 8.02$ kg.

(d) [3 marks]

M1 10 mg/L in 100000 L is 1 kg, so require $x \leq 1$ and $y \leq 1$.

R1 For $t \geq 0$, $x - y = 30e^{-t/2} > 0$ and x is strictly decreasing, so the first safe time is the unique solution of $x(t) = 1$.

A1 Solve $15(e^{-t/10} + e^{-t/2}) = 1$: $t = 27.08069952\ldots \approx 27.1$ h.

(e) [3 marks]

M1 Use simultaneous Euler updates $x_{n+1} = x_n + h(-0.3x_n + 0.2y_n)$ and $y_{n+1} = y_n + h(0.2x_n - 0.3y_n)$.

A1 For $h = 1$: $(x_1, y_1) = (21, 6)$ and $(x_2, y_2) = (15.9, 8.4)$.

R1 The exact $y(2) = 6.762769679\ldots$ kg. The nonnegative Euler value overestimates it by about 24.2% and even exceeds the exact global maximum $8.02488\ldots$ kg. Nonnegativity does not imply accuracy.

(f) [3 marks]

M1 Exchange terms cancel when adding the equations: $S\prime = -0.1S$, $S(0) = 30$, independently of Q.

A1 $S(t) = 30e^{-0.1t}$, so the total half-life is $10 \ln 2 \approx 6.93$ h.

R1 Increasing exchange changes the distribution between reservoirs, but not the total removal rate or half-life under this perfectly mixed model.

(g) [5 marks]

M1 Add and subtract the simultaneous Euler updates.

A1 $S_{n+1} = (1 - 0.1h)S_n$, $D_{n+1} = (1 - 0.5h)D_n$.

A1 Preserving $D_n \geq 0$ for every $D_0 \geq 0$ requires $1 - 0.5h \geq 0$, hence $h \leq 2$.

R1 For $0 < h \leq 2$, the original update coefficients $1 - 0.3h$ and $0.2h$ are nonnegative, so masses remain nonnegative; the difference also stays nonnegative. Thus the largest allowed step is 2 h. If $h > 2$, any $D_0 > 0$ reverses sign after one step.

R1 These conditions preserve qualitative features only. Finite-step truncation error can remain large, as the $h = 1$ computation demonstrates.

End of worked solution book